

OPERATOR NAMING GRAMMAR (Redding Notation)

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June 2026

Operator Naming Grammar (Redding Notation)

This document defines a general naming scheme for operators of the form

$$(\text{generator})_{\text{type}}^{\text{regime}}$$

which serves as a universal annotation style for generative, transformative, or evaluative operators across all systems.

The canonical operator signature is

$$(\text{generator})_{\text{type}}^{\text{regime}}$$

where

Generator : the operator family

Type : the functional role

Regime : the operational mode

More formally,

$$\text{Operator} = G_t^r$$

with

$$G \in \text{Generator Family}, \quad t \in \text{Operator Types}, \quad r \in \text{Operational Regimes}.$$

This makes every operator a member of a three-level taxonomy.

1. Core Pattern

Every operator follows the schema

$$(\text{generator})_{\text{noun/type}}^{\text{verb/adjective}}.$$

Where:

- generator: the base operator (Q , Ξ , T , D , etc.)
- noun/type: operator category or functional role
- verb/adjective: regime, mode, or rule in effect

This pattern is independent of any specific theory.

Level 1 — Generator (Base Family)

Answers the question:

“What family does this belong to?”

Examples:

Q	:	admissibility
Ξ	:	transformation
Γ	:	closure
Δ	:	difference
Σ	:	aggregation
Π	:	projection
Λ	:	lifting
T	:	transport
D	:	dynamics

These are families, not specific operators.

Level 2 — Type (Subscript)

Answers the question:

“What does this operator do?”

Examples:

a	:	admissibility
t	:	transformation
c	:	closure
p	:	projection
e	:	evaluation
n	:	normalization
f	:	filtering
m	:	measurement
g	:	generation

These are roles, not implementations.

Level 3 — Regime (Superscript)

Answers the question:

“Under what rules is it operating?”

Numeric regimes:

1	:	classical
2	:	damped
3	:	hysteretic
4	:	stochastic
5	:	quantum
6	:	nonlinear

Symbolic regimes:

S : structural
 F : field
 D : dynamic
 L : local
 G : global
 A : adaptive

The superscript is a policy selector.

2. Long-Form vs. Short-Form

Long-form (first appearance):

$$Q_{\text{admissibility}}^{\text{classical}}, \quad \Xi_{\text{transformation}}^{\text{structural}}.$$

Short-form (after definition):

$$Q_a^1, \quad \Xi_t^S.$$

Mapping:

$$Q_{\text{admissibility}}^{\text{classical}} \longrightarrow Q_a^1.$$

3. Example: Admissibility Generators

Long-form definitions:

$$Q_{\text{admissibility}}^{\text{classical}} = i, \quad Q_{\text{admissibility}}^{\text{damped}} = i + \delta_{\text{damped}}, \quad Q_{\text{admissibility}}^{\text{hysteretic}} = i + \delta_{\text{hysteretic}}.$$

Short-form equivalents:

$$Q_a^1 = i, \quad Q_a^2 = i + \delta_a^2, \quad Q_a^3 = i + \delta_a^3.$$

4. Example: Xi-Operators

Long-form:

$$\Xi_{\text{transformation}}^{\text{structural}}, \quad \Xi_{\text{transformation}}^{\text{field}}, \quad \Xi_{\text{transformation}}^{\text{dynamic}}.$$

Short-form:

$$\Xi_t^S, \quad \Xi_t^F, \quad \Xi_t^D.$$

5. Summary

All operators follow:

$$(\text{generator})_{\text{type}}^{\text{regime}}.$$

Rules:

- Long-form is used on first introduction.
- Short-form is used thereafter.

- Superscripts index regimes.
- Subscripts denote operator type.

Canonical example:

$$Q_{\text{admissibility}}^{\text{classical}} = Q_a^1 = i.$$

Canonical Form

$$G_t^r$$

Examples:

$$Q_a^1 \quad \text{classical admissibility generator}$$

$$Q_a^3 \quad \text{hysteretic admissibility generator}$$

$$\Xi_t^S \quad \text{structural transformation operator}$$

$$\Pi_p^L \quad \text{local projection operator}$$

$$D_m^4 \quad \text{stochastic measurement operator}$$